14 Partial Derivatives

14.1 Functions of Several Variables

Prerequisite information for this section:

Give a "formal" written definition of a function of one variable ($f(x)$). (It's ok to look it up on the Internet or in your Algebra 1 textbook!)

A rule assigning a variable x to a corresponding value $f(x)$.

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Function of two variables:** [Definition Box above Figure 1.]

  A rule assigning each $(x,y) \in D$ to some $f(x,y)$.

  $f : D \longrightarrow \{f(x,y) \mid (x,y) \in D\}$

- **The Graph of a function of two variables:** [Definition Box above Figure 5.]

  The set of all points $(x,y,z) \in \mathbb{R}^3$ such that $z = f(x,y)$ and $(x,y) \in D$

- **Level curves of a function of two variables:** [Definition Box above Figure 11.]

  The curves with equations $f(x,y) = k$, where $k \in \mathbb{R}$ is some fixed value.

- **Function of three variables:** [Definition given before Example 14.]

  A rule assigning each $(x,y,z) \in D \subseteq \mathbb{R}^3$ a value $f(x,y,z)$.

- **Function of n variables:** [Definition given after Figure 22.]

  A rule that assigns a number $z = f(x_1, \ldots, x_n)$ to each n-tuple $(x_1, \ldots, x_n) \in D \subseteq \mathbb{R}^n$

14.2 Limits and Continuity

Prerequisite information for this section:

1. Give the precise (epsilon-delta) definition of a limit which you "memorized" while taking Calculus 1. (Section 2.4 of your text.)

Let f be defined over an interval I, possibly including a

$\lim_{x \to a} f(x) = L \iff \forall(\varepsilon > 0)(\exists(\delta > 0)(0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon))$

2. Since the function, $z = f(x, y)$, describes a surface over a domain in the xy plane, when we say that $(x, y) \longrightarrow (3, 2)$, in how many ways can (x, y) approach $(3, 2)$?

Infinitely many

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Precise Definition of a Limit of a Function of Two Variables:** [Definition Box 1 of section 14.2.] Let f be defined over an interval D containing points arbitrarily close to (a, b).

$\lim_{(x,y) \to (a,b)} f(x,y) = L \iff \forall \varepsilon > 0 (\exists \delta > 0 (0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \Rightarrow |f(x,y) - L| < \varepsilon))$

- **A Polynomial Function of Two Variables:** [Given below Box #2 in section 14.2.]

A sum of terms $cx^m y^n$, where $c \in \mathbb{R}$ and x, y are non-negative integers

- **A Rational Function of Two Variables:** [Given below Box #2 in section 14.2.]

A ratio of two polynomials.

- **Continuity of a Function of Two Variables:** [Definition Box # 6 in section 14.2.]

A function f is continuous at (a, b) iff

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

- **Precise Definition of a Limit of a Function of n Variables:** [Box # 7 of section 14.2.]

If f is defined over $D \subseteq \mathbb{R}^n$, then

$$\lim_{\vec{x}\to\vec{a}} f(\vec{x}) = L \iff \forall \varepsilon > 0 \left(\exists \delta > 0 \left(0 < |\vec{x} - \vec{a}| < \delta \implies |f(\vec{x}) - L| < \varepsilon\right)\right)$$

where $\vec{x}$ is the position vector $\vec{x} = \langle x_1, \ldots, x_n \rangle$ and $\vec{a}$ is the position vector $\vec{a} = \langle a_1, \ldots, a_n \rangle \in \mathbb{R}^n$.

Topics to learn in this section:

1. While the definitions of limits and continuity for multivariable functions are nearly identical to those of their single variable counterparts, very different behavior can take place in the multivariable case.
2. The idea of points being "close" in $\mathcal{R}^2$ and $\mathcal{R}^3$.

14.3 Partial Derivatives

Prerequisite information for this section:

If you are rusty on any differentiation rules, please have a look at Chapter 3 of your text to brush up. You are expected to be an expert at differentiating functions of one variable.

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Partial Derivatives of a function of two variables:** [Definition Box # 4 in Section 14.3.]

$$f_x(x,y) = \lim_{h \to 0} \frac{f(x+h, y) - f(x,y)}{h}$$

$$f_y(x,y) = \lim_{h \to 0} \frac{f(x, y+h) - f(x,y)}{h}$$

Topics to learn in this section:

1. The meaning of f_x and f_y, both analytically and geometrically.
2. The various notations for f_x and f_y. (Be sure to point out that in the notation $f_x(x,y)$, x is playing two different roles: $f_x(x,y)$ can be written $f_1(x,y)$, where the 1 indicates that the derivative is taken with respect to the first variable.)
3. Higher-order partial derivatives.

14.4 Tangent Planes and Linear Approximations

Prerequisite information for this section:

1. Let $f(x) = 4\cos(x) - \sin(2x)$. Give an equation for the line tangent to this curve at $x = \pi/2$. Use exact values (no calculator approximations). Put your equation in slope-intercept form.

$f'(x) = -4\sin(x) - 2\cos(2x)$ $f(\frac{\pi}{2}) = 0$ $y = -2x + \pi$

$f'(\frac{\pi}{2}) = -4 + 2 = -2$

2. You now have a line, $y = \ldots$. Give calculator approximations (as necessary) to each of the following values, correct (truncated) to the nearest thousandth.

   (a) $f(\pi/2)$ $= 0$

   (b) The y value of your line equation when $x = \pi/2$. (Is this value close to your answer for part (a)? Why?)

   $y = 0$ They are equal because the tangent line was defined to be tangent at $x = \pi/2$

   (c) $f(1.5)$ $= 0.142$

   (d) The y value of your line equation when $x = 1.5$. (Is this value close to your answer for part (b)? Why?)

   $y = 0.142$ This is close because 1.5 is close to $\pi/2$, and a tangent line is a decent approximation for nearby values.

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Tangent Plane:** [Paragraph above Figure 1 in Section 14.4.]

Suppose $z = f(x,y)$ defines a surface with point $P = (x_0, y_0, z_0)$. Let C_1 and C_2 be curves for $y = y_0$ and $x = x_0$ respectively. The tangent plane contains the lines tangent to C_1 and C_2.

- **Linearization of f at (a, b)** (also known as the **linear approximation** or **tangent plane approximation:** [Box # 3 in Section 14.4 with the sentences before and after.]

$f(x,y) \approx L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-a)$

- **Differentiability of a two-dimensional function:** [Definition Box #7 of Section 14.4.]

Let $\Delta z = f(a,b) - f(x,y)$ Let $\varepsilon_1, \varepsilon_2$ be functions of Δx and Δy s.t. $\varepsilon_1, \varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0,0)$
f is differentiable at (a,b) iff

$$\Delta z = f_x(a,b)\Delta x + f_y(a,b)\Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta x$$

Topics to learn in this section:

1. The tangent plane and its analogy with the tangent line.
2. Approximation along the tangent plane and its analogy with approximation along the tangent line.
3. The meaning of differentiability in $\mathcal{R}^2$ and $\mathcal{R}^3$.

In one-variable calculus (calc 1), as soon as we defined the derivative of a function, we immediately started writing down an equation for the line tangent to a function at a point. **Why is the tangent line to a curve at a point such a big deal?**

Now, we are allowing multi-variate functions. Although they are still relatively easy to compute at a point, most real-world problems involve evaluating these functions on an entire grid (2-d), box (3-d), or hyper-dimensional (4-d or higher) region of values. Since this is usually done with matrix arithmetic, it just became extremely important to have a simpler representation of our multi-variate functions. **What should we do?**

The sketch below depicts a surface above the xy plane defined by $z = f(x,y)$. We want to find an equation for the tangent plane to this surface at the point, $P = (x_0, y_0, z_0)$.

Curve C_1: Let x vary. If $y = y_0$, fixed,

Curve C_2: Let y vary. If $x = x_0$, fixed,